SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct: (192260056)

I Parkavi N.D *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	1	2
Concept Mapping Structure	25%	2	1	1
Linking & Relationships	20%	1	2	2
Application & Analysis	20%	1	2	1
Explanation & Clarity	15%	1	1	1

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

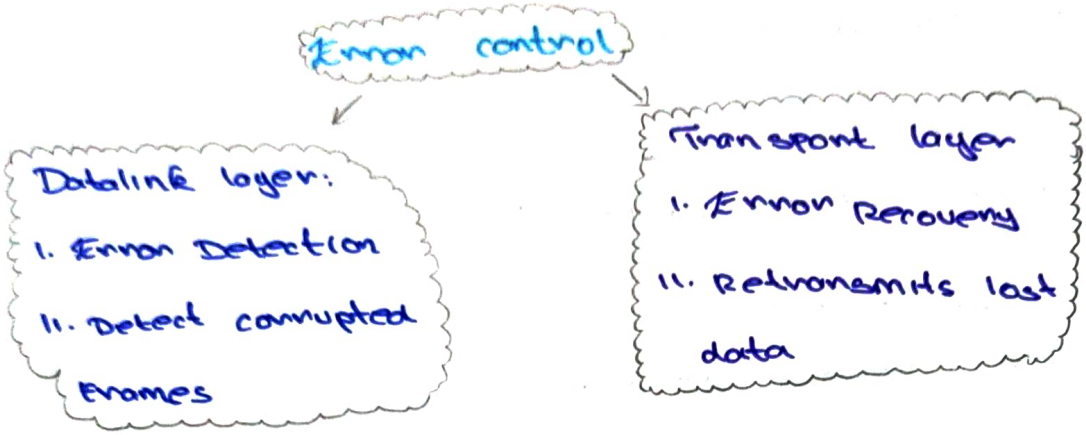
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

CO-1 At-2

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(4) Error control in Data transmission

concept Map:



Analysis:

1. The Data link detects errors in frames using methods such as CRC

2. The transport layer, using TCP, detects missing or incorrectly received data and heques retransmission when necessary

3. Both layers & work together to ensures reducing data loss and corruption

4. Conclusion

Errors detection and recovery helps provide reliable data transmission

(5) concept Map for web Browsing using osi layers

concept Map:

7. Application

HTTP /HTTPS - web request

↓

6. presentation

Data format and encryption

↓

5. session

Manages communication sessions

↓

4. Transport

TCP - Reliable delivery

↓

3. Network

IP - Routing Packets

↓

2. Data link

Frames and Mac addresses

↓

1. physical

Signal through cable or wifi

Application of Failures:

* Application layer: Incorrect URL or browser error prevents occur to website

* Presentation layer: Encryption or data format problems may prevent the page from loading correctly

*Session layer: session failure may interrupt communication

*Transport layer: TCP problems can cause delays

*Network layer: IP or routing problems prevent packets from reaching this server

*Data link layer: wifi or Mac address problem disrupt local network communication

*physical layer: A damaged cable or weak wifi original causes connection Failure

*conclusion: The OSI model helps identify which layer is responsible for web-browsing problem

6) Network Trouble shooting using The OSI Model

concept Map:

Network issue

↓

physical layer

check cables, power and wifi

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Data link layer

check Network adapter and Mac connectivity

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Network layer

check IP address, gate way and routing

Transport layer

check Tcp /Udp ports and connection

Application layer

check browser, DNS - related settings and services

Analysis:

1. Sart with physical layer by checking cables and wifi
2. check the Data link layer for adapter and local connecting problem
3. verify this ip address, default gateway and routing at the network layer
4) check the port and Tcp/Udp connection at Transport layer

How the concept map helps

* It provide a systematic trouble shooting order
* It helps isolate the source of the problem
* It makes network Fault identification port and easier

Conclusion:

The osi model concept map simplifies Trouble shooting by dividing network communication into layers and helping locate the Faculty compound